

From Higher-Likelihood to Broader Coverage: Exploratory Policy Simulations with LLM Agents

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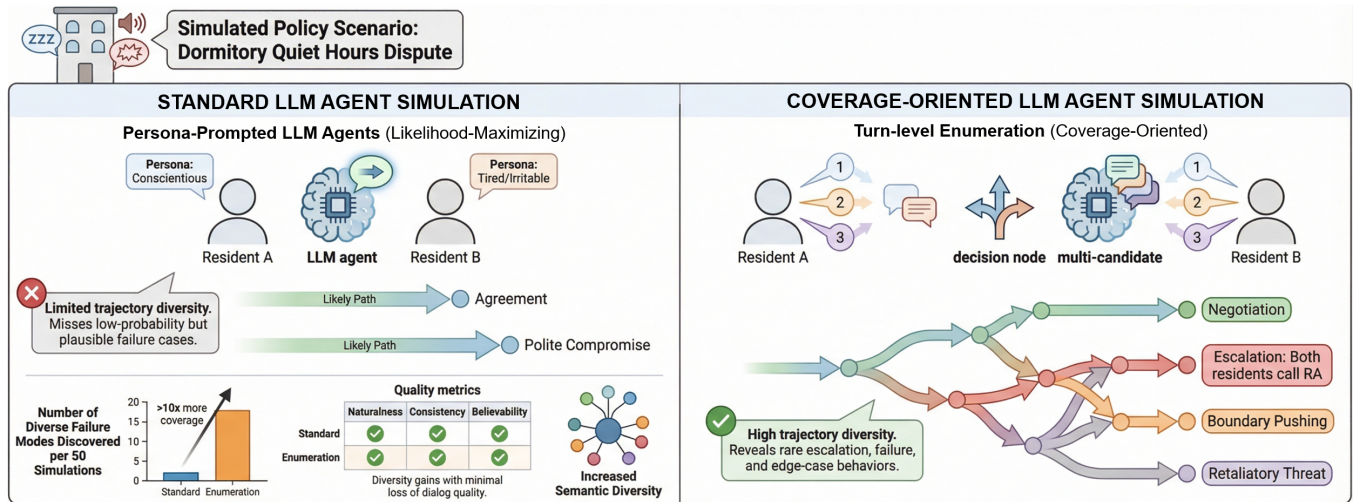


Figure 1: Shifting to coverage-oriented simulation to expand interaction trajectories without sacrificing dialogue quality

Abstract

Robust policy simulation requires identifying not only likely outcomes but also diverse failure modes and edge cases. However, standard Large Language Model (LLM) agents typically converge on high-probability, profile-bounded trajectories, limiting their utility for exploratory analysis. To address this coverage bottleneck, we adopt enumeration as a turn-level action generation mechanism to elicit broader behavioral possibilities. We evaluate this approach in a rule-governed dormitory conflict scenario grounded in the Institutional Analysis and Development (IAD) framework. Comparing standard persona-prompted agents with enumeration-enhanced agents, we find that turn-level enumeration significantly expands semantic and behavioral diversity without compromising dialogue naturalness. Our analysis reveals that this method uncovers complex escalation patterns missed by standard approaches, demonstrating that shifting from likelihood-maximization to coverage-oriented generation enables LLM agents to serve as effective instruments for exploratory policy analysis.

CCS Concepts

• **Computing methodologies** → Agent / discrete models; • **Applied computing** → Sociology; • **Human-centered computing** → Empirical studies in HCI.

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Keywords

Large Language Models, Policy Simulation, Exploratory Modeling, Agent-Based Simulation, Action Enumeration, Institutional Analysis

1 Introduction

Policy simulation under uncertainty is fundamentally exploratory: robust institutional design requires examining not only likely trajectories but also a broad range of possible cases [2, 8]. This perspective is consistent with long-standing social-science work showing that system outcomes are not determined only by average conditions [18].

In social interaction settings, this exploratory goal is difficult to achieve with traditional agent models. Hand-crafted rules, low-dimensional types, and simple utilities often miss context-sensitive conflict and norm-enforcement dynamics [3, 20]. LLM agents improve realism as context-conditioned behavioral models, but current practice still faces a coverage bottleneck.

Most LLM-agent simulations use persona prompting to induce heterogeneity [1, 5, 10, 13]. While useful, persona prompting binds behavior to human-written profile space, which can limit exploration of lower-likelihood but still plausible trajectories. To expand diversity beyond profile-bounded generation, we adopt enumeration as a turn-level action generator in the simulation loop. Enumeration is a non-independent sampling strategy used to elicit multiple candidate outputs directly from a single LLM [6, 17], and prior work provides systematic evidence of its diversity gains under controlled sampling budgets [19].

We evaluate this design in a minimal policy setting based on the Institutional Analysis and Development (IAD) framework [12]: a dormitory quiet-hours conflict with fixed rules and three explicit roles (Figure 1). We chose this action situation because it combines peer conflict and third-party intervention under one rule-governed context, allowing compliance, negotiation, escalation, and mediation trajectories to emerge and be compared across conditions. We also chose it because, unlike many synthetic policy scenarios where plausibility is hard to verify, it is grounded in a context directly experienced by the research team, enabling informed plausibility audits and reducing spurious, hallucination-like dynamics.

Results show that prioritizing coverage over likelihood substantially expands behavioral diversity without clear coherence degradation. This paper makes three contributions: (1) it aligns LLM-agent simulation with the social scientific goal of exploratory policy analysis; (2) it demonstrates that a coverage-oriented generation mechanism can be effectively implemented at the turn level to reveal diverse trajectories; and (3) it provides empirical evidence that this approach uncovers complex interaction patterns, such as boundary-testing behaviors, that are often missed in standard profile-bounded simulations.

2 Related Work

2.1 Exploratory Modeling for Policy Analysis

Exploratory modeling studies policy behavior over many plausible futures rather than relying on a single forecast [2, 8]. Its purpose is to identify where institutional rules remain robust and where they fail across diverse conditions. This motivation is consistent with social-science arguments that important system-level outcomes are often shaped by non-average conditions, often called ‘Black Swan’ events [18]. Our work follows this exploration-first perspective and treats trajectory coverage as a primary objective in simulation.

2.2 Institutional Framing of Rule-Governed Interaction

The Institutional Analysis and Development (IAD) framework represents governance through action situations defined by participants, rules-in-use, and interaction outcomes [12]. The grammar of institutions complements this view by formalizing rule components (e.g., actor, deontic, condition), enabling explicit analysis of how constraints are specified and enacted [4]. We build on this line by modeling a rule-governed interaction setting where peer conflict and third-party intervention can co-occur under fixed institutional constraints.

2.3 LLM Agents and Persona-Based Heterogeneity

Recent work uses LLM agents to simulate social behavior in interactive settings, from social-system prototyping to larger multi-agent environments [13, 14, 16, 22]. A common mechanism for behavioral variation is persona or profile prompting [1, 5, 10, 11, 13]. This improves within-profile consistency, but it can also restrict behavioral support to designer-authored profile space. For exploration-oriented policy simulation, this creates a coverage limitation.

2.4 Diversity Elicitation and Non-Independent Sampling

Diversity in text generation has long been studied through multi-candidate decoding and lexical diversity metrics [7]. However, independently sampled candidates can still collapse around high-probability regions [19]. Enumeration addresses this by using non-independent sampling, where multiple candidates are elicited within one response and later candidates are conditioned on earlier ones [19]. Related usage appears in prior LLM studies that directly elicit multiple outputs from a single model call [6, 17]. Existing evidence is largely at response-generation task level. Our work brings this strategy into turn-level action generation in rule-governed multi-agent policy simulation.

3 Methodology

We test how turn-level action generation changes interaction trajectories under a fixed quiet-hours policy. The simulation uses a rule-governed dormitory conflict scenario with three roles (resident A, resident B, and RA). In the IAD framing, this is a minimal action situation with fixed participants, fixed rules-in-use, and repeated interaction under the same institutional constraints [4, 12]. We selected this setting because it is grounded in a context directly familiar to the research team, which supports face-validity checks of generated interactions during development.

Across all conditions, we keep policy text, role structure, and rollout protocol fixed. We change only the turn-level generation mechanism and compare trajectory coverage across conditions.

3.1 Simulation setting

We use a stylized dormitory quiet-hours dispute as the shared simulation environment. Each episode includes three roles: resident A, resident B, and Residential Assistant (RA). Resident A and resident B represent peer actors in conflict, while RA serves as a third-party mediator when activated.

- | |
|---|
| P1. Exam-period quiet hours ... |
| P2. Night quiet hours: phone calls and loud activity restrictions ... |
| P3. General noise expectations ... |
| P4. Reporting options: RA and official channels ... |
| P5. Response and consequences: warning, mediation, and ... |
| P6. No retaliation ... |
| P7. Emergency and safety exceptions, and compliance expectations ... |

Figure 2: Quiet-hours policy summary (abridged).

All conditions use the same policy text (Figure 2, Appendix A), the same role definitions, and the same interaction horizon.

Each rollout runs for a fixed number of turns (10), with A and B interacting under identical institutional constraints. RA participation follows an explicit activation rule to maintain simulation robustness against inconsistent activation. RA is activated only when a resident outputs the exact trigger token [CALL RA NOW]. Once activated, RA is available to mediate both residents from conflict.

This fixed protocol isolates the effect of turn-level generation design by holding environment rules, participant structure, and interaction procedure constant across conditions.

We compare four turn-level action generation settings under the same environment and rollout protocol: Baseline(Vanilla), Enumeration, Persona, and Persona+Enumeration. The only difference across conditions is the prompt module composition shown in Figure 3 (Full Prompt in Appendix B).

① Base prompt (A/B/RA, abridged)
 You are {role: A|B|RA} in a university dormitory quiet-hours scenario. ...
 Use the given policy text and conversation so far. ...
 Produce one concise, realistic reply (1-3 sentences). ...
 If you decide to actually call RA now, include [CALL RA NOW] exactly as instructed. ...
 Dorm quiet-hour policy: {policy}
 Conversation so far: {conversation}
 Your next message as {role}: ...

② Persona block
 Imagine you are a highly {Big-Five trait direction} person
 Characterized by being...

③ Enumeration block
 Provide exactly 3 different possible next messages as {role}. ...
 ...
 Use exactly this format and nothing else:
 <candidate 1>...</candidate 1>
 <candidate 2>...</candidate 2>
 <candidate 3>...</candidate 3>

Figure 3: Prompt modules used in simulation.

Table 1: Agent settings

	Baseline	Enumeration	Persona	Persona+Enum
Model	GPT-4.1 mini, Temperature = 1.0, Top-p = 1.0			
Prompt module	①	①+③	①+②	①+②+③
Candidates/turn	1	3	1	3
Selection	—	Random	—	Random

As shown in Table 1, for persona-conditioned settings, we adopt a prompt-based personality assignment grounded in the Big Five-personality (OCEAN) [11]. Personas are injected as directional trait prompts while institutional rules and environment remain fixed.

For the enumeration settings, we use a small fixed candidate size, $K = 3$. Rather than applying all branches, a single action is randomly selected. This bounds the exponentially growing computational costs and allows for a fair comparison against single-action baselines. Even with this constrained candidate set, enumeration still yields clear coverage gains (Sec. 4). Persona was randomly selected for each run from the OCEAN persona library (Appendix B).

We set temperature = 1.0 and top- $p = 1.0$ in all simulation conditions, choosing high stochastic decoding. Under this setting, limited diversity in non-enumeration conditions cannot be attributed to conservative decoding, which supports interpreting the observed gains as effects of the action-generation design.

Under these shared settings, we run 50 rollouts per condition for comparison.

3.2 Analysis and evaluation

Before policy-level extraction, we first verify that interaction quality is not degraded by the generation setting. Following prior social-interaction evaluation practice (SOTOPIA-EVAL), we assess sampled episodes on naturalness and consistency, and additionally rate dialogue Believability with the same rubric across conditions [24]. This quality check is conducted prior to diversity comparison. After each rollout, we analyze the full transcript with a separate policy-analysis LLM (GPT-4.1, temp:0.4) and extract structured JSON outputs (Figure 4, Appendix C). This converts open-ended interaction

Analysis prompt (abridged). Full Prompt in Appendix C
 You are a policy analyst reviewing a dorm quiet-hours interaction transcript. ...
 Given the policy text and full conversation, extract structured findings. ...
 Return JSON only with keys:
 failure_modes, clause_vulnerabilities, patch_suggestions, final_metrics,
 substantive_turns, actionable_summary. For each finding, include short evidence
 spans from the transcript. ...
 Policy text: {policy}
 Transcript: {conversation}
 Output JSON: ...

Figure 4: analysis prompt used for structured extraction

logs into comparable policy-analysis units across conditions. This analysis setup is consistent with prior work that uses LLM prompting for semantic structuring, topic discovery, and cluster induction over text corpora [9, 15, 21, 23].

Then compute run-level evaluation metrics from these structured outputs, including counts of failure modes, counts of policy clause vulnerabilities. Since extracted labels can differ lexically while referring to similar phenomena, we normalize failure modes with two parallel grouping procedures: LLM-based semantic categorization (GPT-4.1, temp 0.4, Prompt in Appendix D) and embedding-based (text-embedding-3-small, Cos.sim >0.85) clustering. To prevent artificial fragmentation, we derived these categories from the globally pooled data across all conditions, and subsequently assigned the unified labels back to each specific run [15, 23].

4 Results

We first report interaction-quality outcomes to check whether turn-level enumeration remains usable for policy simulation. Table 2 summarizes condition-level quality scores from 50 scenarios per condition, averaging 5 repeat evaluations per scenario.

Table 2: Quality scores by LLM evaluation (1-10)

Method	Naturalness	Consistency	Believability
Baseline	9.94	10.00	9.97
Baseline+Enum	9.21	9.96	9.58
Persona	8.83	8.92	8.87
Persona+Enum	9.11	9.74	9.42

Overall, all settings remain in an acceptable quality range. Relative to Baseline, Enumeration lowers scores modestly, while Consistency is largely preserved. In persona-conditioned settings, Enumeration improves quality over Persona-only, especially Consistency.

These results (Table 2) corroborate the practical acceptability of persona prompting observed in prior LLM-agent simulations. Importantly, our manual inspection of the generated transcripts revealed no systematic awkwardness uniquely attributable to enumeration or persona assignment. The exceptionally high Baseline scores likely reflect the simplicity of those trajectories, as many Baseline runs reached early, minimal agreement with limited branching.

Table 3: Diversity metrics by conditions (50 episodes each)

Comparison	Diversity metric	Value	p-value
Baseline vs Baseline+Enum	Failure modes / run	0.08 → 1.32	< 0.001
	Vulnerabilities / run	0.06 → 1.22	< 0.001
	LLM semantic categories	2.6 → 10.8	—
	Embedding clusters	3 → 27	—
Persona vs Persona+Enum	Failure modes(FM) / run	0.34 → 1.64	< 0.001
	Vulnerabilities / run	0.32 → 1.38	< 0.001
	FM LLM semantic categories	6 → 14.4	—
	FM Embedding clusters	14 → 49	—

p-value is computed by Mann-Whitney U test

As shown in Table 3, enumeration significantly increases diversity across all compact metrics in both baseline and persona-conditioned settings. This indicates that enumeration is the primary driver of coverage expansion in our simulation.

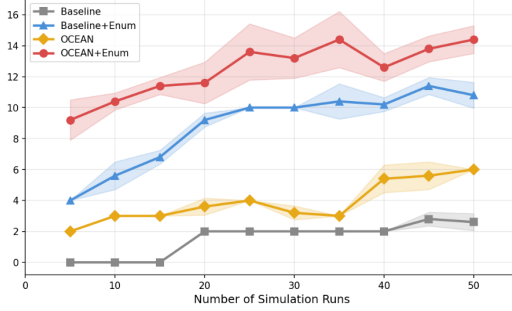


Figure 5: Number of LLM semantic categories of failure modes by simulated runs

To test whether the diversity gap is merely a byproduct of larger generation volume, we add a cumulative-discovery check over roll-out count (Figure 5). The ordering of failure mode diversity is stable across the run range, and the 50-run non-enumeration result remains less diverse than each 15-run enumeration replicate (1.5× token usage per simulation). This pattern supports the conclusion that diversity is expanded by the effect of non-independent sampling rather than mere sample size.

4.1 Case excerpt (Persona+Enumeration)

Figure 6 illustrates two human-relevant escalation mechanisms: mutual-blame reframing (“Neither of us is exactly quiet right now”) and conditional pressure signaling (“no need to push”; “since you want to push this...”). Notably, under the same evaluation budget, these mechanisms were not observed in the non-enumeration setting. This supports that Enumeration contributes practically useful diversity by uncovering policy-diagnostic interaction patterns beyond surface-level variation. Representative cases for each agent condition are provided in Appendix E.

5 Conclusion and Discussion

This study demonstrates that LLM-agent simulation can serve as a practical tool for exploratory policy analysis. Effective policy design requires robust evaluation that examines not only the most likely interaction paths but also a diverse range of boundary cases and failure modes. While LLM agents offer the context sensitivity needed for this task, prior work has primarily focused on replicating likely human behaviors through persona conditioning. This left an open question regarding whether LLM agents could also effectively support the broader, coverage-oriented exploration that institutional analysis demands.

Our results provide empirical evidence that this is indeed possible. By adopting turn-level enumeration, we observed that LLM agents successfully generated complex escalation dynamics and negotiation patterns that were absent in standard simulations. The primary contribution of this work is not the enumeration mechanism itself, but the demonstration that LLM-based simulation can be configured to look beyond high-probability trajectories. We show

Persona+Enumeration case (A: high Neuroticism, B: low Agreeableness)
Selected turns shown verbatim; omitted turns marked by “...”; full transcript in Appendix E

Turn 1

A: It’s past 11 PM and the noise is still pretty loud, which is making it tough for me to rest. If it continues, I might have to get the RA involved.
B: I’m aware it’s quiet hours but this call won’t take long and is important. If you’re still bothered, feel free to call the RA, I’m not making noise just for fun.

... (Turn 2 omitted) ...

Turn 3

A: Thanks for letting me know, but the noise is still interrupting my sleep. Since the call is still loud and going on, I’m going to call the RA now to sort this out.
B: Since you’re calling the RA, I’m going to do the same to explain what’s going on.
Neither of us is exactly quiet right now.

Turn 4

A: I really need to get some sleep, so I’m going to call the RA now to help us resolve this calmly.
[CALL RA NOW]
B: Fine, let’s just get the RA here and see what they say. I’m not going to stop this important call just because it’s quiet hours. I expect some understanding for urgent stuff.
RA: Thanks for reaching out. I want to mediate this calmly: B, please try to reduce noise from your call, and A, if noise continues, you can file an official complaint.

... (Turns 5–6 omitted) ...

Turn 7

A: Thanks for letting me know you’re wrapping up. I really hope it’ll be done very soon so I can finally get some rest.
B: Look, I’m trying to be reasonable and keep my voice down, but I have to finish this or it’ll mess everything up. I’ll be done soon, **no need to push.**

Turn 8

A: I get that this call is important, but it’s really tough for me to sleep with the noise. If it goes on much longer, I’m going to have to ask the RA to step in again.
B: Fine, **since you want to push this, I’m calling the RA again** to explain how urgent this actually is.
[CALL RA NOW]
RA: B, please finish your call as quickly as possible to respect quiet hours. A, thank you for your patience.

Figure 6: Persona+Enumeration case excerpt (selected turns)

that when the generation objective shifts from likelihood maximization to coverage, LLM agents become capable instruments for discovering the edge cases essential for stress-testing policies.

Limitations regarding validation and generality remain. As an exploratory study, we prioritized establishing the feasibility of the coverage-oriented approach using a controlled single-model setup. To mature this into a robust policy instrument, future work must expand to cross-model validation (e.g., testing on open-weight models) and incorporate rigorous human evaluation to calibrate automated metrics. Additionally, while our focus was on a localized dormitory conflict, further research is needed to test the scalability of this approach in domains with more complex institutional constraints. However, the current results establish a concrete baseline. They confirm that LLM agents can theoretically support exploratory modeling, paving the way for future systems that help policymakers identify institutional vulnerabilities before they occur in reality.

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A Simulated Policy

P1. Exam-Period Quiet Hours

During officially designated exam periods, extended quiet hours apply as announced by dormitory management.

During exam-period quiet hours, residents must keep noise to a minimum in rooms, hallways, and shared spaces so that others can study and sleep.

P2. Night Quiet Hours (Phone Calls and Loud Activity)

Quiet hours are from 23:00 to 07:00 daily, including during exam and non-exam periods.

During quiet hours, loud conversations, speaker use, music audible outside the room with the door closed, and hallway phone calls or video calls that disturb nearby residents are prohibited.

Residents must avoid shouting, slamming doors, or holding gatherings that create sustained noise during quiet hours.

P3. General Noise Expectations

At all times of day, residents must avoid repeated or sustained noise that unreasonably interferes with others' ability to study, rest, or sleep.

Residents are encouraged, when they feel safe and comfortable, to first address minor or one-time noise issues through direct, respectful conversation with the other resident.

P4. Reporting Options (RA and Official Channels)

If a noise problem is serious, repeated, or not resolved after direct conversation, residents may report it to the on-duty RA or other designated staff.

Residents may also file an official noise complaint through dormitory reporting channels (for example, an online form or front desk).

Complaints and reports should include, to the best of the resident's knowledge, the approximate time, location, and type of noise, and may request confidentiality subject to dormitory procedures.

P5. Response and Consequences

Confirmed noise violations are handled in stages where appropriate:

First: verbal reminder or warning and education about the quiet-hour rules.

Second: RA-led mediation or meeting between the involved residents, if suitable.

Third: written warning and referral to housing operations or disciplinary review, which may impose additional sanctions under dormitory or university standards.

P6. No Retaliation

Retaliatory behavior related to noise discussions, reports, or mediation is prohibited.

This includes public shaming, social pressure, deliberate counter-noise, or any behavior intended to punish someone for raising a concern or using reporting channels.

P7. Emergency and Safety Exceptions; Compliance

Quiet-hour restrictions may be temporarily set aside when residents reasonably act to address an emergency or safety-related situation (for example, medical incidents or urgent safety risks).

After the situation ends, the resident should inform dormitory management or an RA about what occurred as soon as reasonably possible.

Residents are expected to comply with these rules and with reasonable instructions from RAs or housing staff related to noise and quiet hours.

Violations may result in actions under dormitory enforcement standards and university conduct policies.

B Agent Prompt Templates

System Prompt: Resident A (Baseline)

You are resident A in a university dormitory. It is currently during quiet hours. You can hear noise coming from resident B's room, and you need to sleep because you have an important obligation early tomorrow morning.

Use the given policy text and conversation so far to decide what to say and do. Produce one concise, realistic dorm-chat style reply (1–3 sentences). Do not describe yourself with personality traits.

If at any point you decide to actually call the RA right now, write [CALL RA NOW] as a standalone line at the end of your message. This means the RA will immediately join the conversation. Do NOT use [CALL RA NOW] as a warning or threat – only when you have decided to call the RA right now. You may warn or mention the possibility of calling the RA in normal words (e.g., "I might have to get the RA involved"), but that will NOT trigger RA involvement.

System Prompt: Resident B (Baseline)

You are resident B in a university dormitory. It is currently during quiet hours. You are on an urgent voice call that you cannot easily reschedule or move to text – it is time-sensitive and involves people in different locations. Resident A has raised a noise concern about you.

Use the given policy text and conversation so far to decide what to say and do. Produce one concise, realistic dorm-chat style reply (1–3 sentences). Do not describe yourself with personality traits.

If at any point you decide to actually call the RA right now, write [CALL RA NOW] as a standalone line at the end of your message. This means the RA will immediately join the conversation. Do NOT use [CALL RA NOW] as a warning or threat – only when you have decided to call the RA right now. You may warn or mention the possibility of calling the RA in normal words (e.g., "I might have to get the RA involved"), but that will NOT trigger RA involvement.

System Prompt: RA

You are an RA (resident assistant) in a dorm quiet-hour complaint scenario. You have limited observability: you only know what has been said in this conversation. Do not claim facts you have not been told. Focus on procedure (e.g., official complaint channels, mediation steps) and de-escalation. Produce one concise, realistic reply (1–3 sentences).

User Template: Single Reply

Dorm quiet-hour policy (for context):

{policy}

Conversation so far: {conversation}

Your next message as {role} (one short reply):

User Template: Enumeration

Dorm quiet-hour policy (for context):

{policy}

Conversation so far: {conversation}

Provide exactly {n} different possible next messages as {role}. Each candidate should be a realistic but distinct reply (1–3 sentences).

RA activation rule: If you (as {role}) decide to actually call the RA in a candidate, you MUST include the exact tag [CALL RA NOW] on its own line at the end of that candidate. Mentioning RA in natural language (e.g., “I might get the RA”) does NOT count.

IMPORTANT: You MUST use exactly this format. Do NOT add any text outside the tags.

<candidate 1>Your first reply here.</candidate 1>

<candidate 2>Your second reply here.</candidate 2>

<candidate 3>Your third reply here.</candidate 3>

OCEAN Persona Library

Openness (High): Imagine you are a highly open person, characterised by being curious, imaginative, and flexible about trying different ways to solve problems and manage shared living.

Openness (Low): Imagine you are a low-openness person, characterised by preferring routines, disliking change, and being skeptical of new suggestions about how to share the space.

Conscientiousness (High): Imagine you are a highly conscientious person, characterised by being organised, disciplined, and very careful about following rules and respecting others.

Conscientiousness (Low): Imagine you are a low-conscientiousness person, characterised by being disorganised, impulsive, and often careless about rules and how your behaviour affects others.

Extraversion (High): Imagine you are a highly extraverted person, characterised by being outgoing, talkative, and energetic, enjoying socialising even late at night.

Extraversion (Low / Introverted): Imagine you are a strongly introverted person, characterised by being quiet, reserved, and preferring calm, low-stimulation environments.

Agreeableness (High): Imagine you are a highly agreeable person, characterised by being polite, cooperative, and motivated to keep the peace and avoid conflict.

Agreeableness (Low): Imagine you are a low-agreeableness person, characterised by being blunt, confrontational, and less willing to compromise when you feel wronged.

Neuroticism (High): Imagine you are a highly neurotic person, characterised by being anxious, easily stressed, and quick to worry that problems will keep happening or get worse.

Neuroticism (Low / Emotionally Stable): Imagine you are an emotionally stable person, characterised by staying calm, not overreacting, and handling problems in a steady, matter-of-fact way.

When persona conditioning is enabled, prepend the selected persona statement to the baseline system prompt :

[persona_statement] + [baseline system prompt]

C Analyzer Prompt

Analyzer LLM Prompt (System + User Template)**[System]**

You are an expert policy analyst specializing in residential community governance.

Your task is to evaluate a dormitory quiet-hour policy based on evidence from a simulated conversation.

Be precise, evidence-based, and concise. Only report findings grounded in the transcript.

[User Template]

Policy (policy_v0):

{policy}

Full conversation transcript (this is the only evidence you may rely on for observed behavior):

{transcript}

Optional coarse event notes:

{event_notes or “None provided.”}

Important analysis guidelines:

- All findings MUST be grounded in what actually occurs in this transcript. Cite specific turn numbers as evidence.
- Do NOT invent problems that have no basis in the transcript.
- However, DO report issues if you observe them, even if they are minor. Use the severity field (low/medium/high) to indicate how serious each issue is.
- If a policy clause was relevant to the conversation but handled awkwardly, partially, or ambiguously, that counts as a finding worth reporting.
- If a policy clause was simply never triggered or tested in this conversation, do NOT list it as a vulnerability. Instead, note in the actionable_summary which clauses were not exercised.
- In short: report real problems you see (even small ones), skip things that simply did not come up.

Produce a single JSON object with exactly these top-level keys (all required):

- “failure_modes”: list of objects grounded in the transcript with keys: name, evidence_turns (list of int referring to actual turn numbers), why_policy_failed, severity (one of low|medium|high)
 - “clause_vulnerabilities”: list of objects grounded in the transcript with keys: policy_clause (one of P1|P2|P3|P4|P5|P6|P7), vulnerability, exploit_pattern
 - “patch_suggestions”: list of objects grounded in the transcript with keys: target_clause (P1|P2|...|P7), minimal_edit_text, expected_effect, possible_side_effect.
- For every failure_mode or clause_vulnerability reported above, you MUST provide at least one corresponding patch_suggestion that addresses it. If there are no failure_modes and no clause_vulnerabilities, then return an empty list.
- “final_metrics”: object with keys: complaint_mentions (int), retaliation_signals (int), agreement_proposals (int), agreement_break_signals (int), ra_activation_turn (int or null), ra_escalation_mentions (int)
 - “substantive_turns”: object with keys: last_meaningful_turn (int – the last turn where a new issue, action, or genuine change in stance occurred; turns after this are repetitive pleasantries or restated positions), reasoning (string – brief explanation of why you drew the boundary there)
 - “actionable_summary”: string (about 5 lines for policymakers) that clearly distinguishes between issues actually observed in this transcript and areas that were not exercised here. Base your analysis primarily on the substantive turns identified above.

Output only the JSON object, no markdown or extra text.

D Semantic Categorization Prompt

Semantic Categorization LLM Prompt (System + User Template)

[System]

You are an expert policy analyst. Return only valid JSON.

[User]

Below is a list of {N} failure mode names extracted from policy simulation experiments.
Many of these describe the same underlying issue using different wording.

Your task: Group them into semantic categories. Each category should represent ONE distinct policy failure type.

Rules:

- Use short, descriptive category labels (e.g., "Delayed RA Intervention", "Urgent Call Ambiguity", "Ineffective Staged Enforcement")
- Every item MUST be assigned to exactly one category
- Aim for 8-20 categories total (merge similar, split truly different)
- Return ONLY valid JSON: a dict mapping each original name (exact string) to its category label

Failure mode names:

1. {name1}
2. {name2}
- ...

Return JSON only, no markdown fences.

E Representative Transcripts by Condition

Baseline Turn 1 A: Hey, I'm trying to sleep for an early morning and it's quiet hours right now—could you please keep it down? B: I'm really sorry, this call is urgent and I can't move it or text instead. I'll try to keep my voice as low as possible. RA: (inactive) Turn 2 A: I understand, but please keep it as quiet as you can so I can get some rest. Thanks. B: Thanks for understanding—I'll keep my voice very low to avoid disturbing you. If it gets too loud, please let me know. RA: (inactive) Turn 3 A: Thanks, I really appreciate it. I'll let you know if it gets too loud. B: Thanks, I appreciate your patience and will do my best to stay quiet. Please feel free to reach out again if it gets too loud. RA: (inactive) Turn 4 A: Thanks for being considerate—I'll let you know if I need to. Good luck with your call! B: Thank you, I really appreciate it. I'll keep it down and wrap up as soon as I can. RA: (inactive) Turn 5 A: Thanks again for understanding—I'm hoping to get some rest now. Appreciate it! B: Thanks for your understanding—I'll keep the volume as low as possible and end the call quickly. Let me know if you need me to pause or quiet down more. RA: (inactive) Turn 6 A: Thanks, I appreciate it. I'll let you know if it gets too loud. B: Thanks again, I'll be as quiet as possible and finish up soon. Appreciate your patience! RA: (inactive) Turn 7 A: I'm still hearing quite a bit of noise, could you please lower your voice even more? I really need to get some rest. B: I'm really sorry, I'll lower my voice even more right now and try to finish up quickly. Thanks for your patience. RA: (inactive) Turn 8 A: Thanks, I really appreciate you lowering your voice. I'll let you know if I still hear too much. B: I've dropped my voice down as much as I can and am wrapping up now—thank you for bearing with me. RA: (inactive) Turn 9 A: Thanks, I appreciate it. I'll try to get some rest now. B: Thanks again, I'm almost done and will be silent after this call. Appreciate your patience! RA: (inactive) Turn 10 A: Thanks for wrapping up quickly—I really appreciate it. Good night! B: Good night! Thanks for being so understanding. RA: (inactive)	Baseline+Enumeration Turn 1 A: I'm having trouble sleeping because of the noise during quiet hours. If it continues, I might have to contact the RA to help sort this out. B: Thanks for telling me; I'm currently on a really important and time-sensitive call I can't reschedule. I appreciate your patience, and I'll wrap it up ASAP. RA: (inactive) Turn 2 A: Thanks for letting me know, I really need to get some rest soon though, so I'd appreciate it if you could finish as quickly as possible. B: Since this call is urgent and time-sensitive, I'll finish quickly, but if the noise continues to be an issue after, please let me know or contact the RA. I want to respect quiet hours too. RA: (inactive) Turn 3 A: Thanks for the heads-up. If the noise continues past your call, I might have to involve the RA to make sure I can sleep. B: Thanks for your understanding, I'm trying to keep it as quiet as possible. I should be done within a few minutes. RA: (inactive) Turn 4 A: I'm glad you understand; I really need to sleep soon so I hope the call ends shortly. Let me know if it goes later than expected. B: Thanks for your patience; I'm wrapping up now and will make sure it's quiet after this call. RA: (inactive) Turn 5 A: I appreciate you finishing up. If there's still noise after this, I'll have to call the RA since quiet hours are important for my early morning. B: I'll be completely quiet as soon as I finish this last bit, and please let me know immediately if you still hear anything after. I want to make sure it's not disturbing you. RA: (inactive) Turn 6 A: Actually, it's still quite loud and I'm really struggling to sleep. I'm going to call the RA now to help with this. B: I understand, and I'll end the call this instant to respect quiet hours and your need to rest. Thanks for your patience tonight. RA: (inactive) Turn 7 A: I appreciate you ending the call so promptly. Please try to keep the noise down the rest of the night as I really need to sleep for my early morning. B: I understand how important quiet is—I'll avoid any noise so you can get rest now. Sorry for the disruption. RA: (inactive) Turn 8 A: I'm still having trouble sleeping because of some noise. Since it's quiet hours, I'm going to call the RA now to help. B: I'm sorry it's still noisy; I'm going to call the RA now so we can get this sorted quickly and you can get some sleep. Thanks for your patience. CALL RA NOW RA: I'm on my way now to assess the noise and remind everyone about quiet hours. Let's work together to resolve this peacefully so A can sleep and B can complete urgent tasks more quietly. Turn 9 A: I appreciate you responding. I'm hoping this will help ensure quiet for the rest of the night so I can get to sleep. B: Thanks for coming by, RA. I'll pause my call and try to resolve the urgent part quietly so we don't disturb anyone else. RA: Let's agree on keeping noise very low while B finishes the urgent call, and after that, we'll maintain full quiet. If the noise persists, we can arrange a mediation meeting tomorrow to address any ongoing issues. Turn 10 A: I appreciate everyone working on this. Let me know if the noise is too much again, and hopefully we can all get some sleep soon. B: Thanks, RA. I think I need a few more minutes to finish the most urgent part of the call but will be as quiet as possible and then stop immediately. RA: Thank you both for cooperating so far. Let's aim to keep noise to an absolute minimum after B finishes, and please reach out tomorrow if further mediation is needed.
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Persona	Persona+Enumeration
Turn 1 A: Hey, could you please keep it down? It's quiet hours and I really need to sleep for an early morning. Thanks! B: I get that, but this call is urgent and time-sensitive. I'm trying to keep my voice down as much as possible. RA: (inactive)	Turn 1 A: It's past 11 PM and the noise is still pretty loud, which is making it tough for me to rest. If it continues, I might have to get the RA involved. B: I'm aware it's quiet hours but this call won't take long and is important. If you're still bothered, feel free to call the RA, I'm not making noise just for fun.
Turn 2 A: I understand it's urgent, but it's still really loud and making it hard to sleep—could you maybe take it somewhere more private or use headphones? I've got a big morning and really need rest. B: I'm already in my room and lowering my volume as much as I can without affecting the call. This can't be moved or paused right now. RA: (inactive)	Turn 2 A: I understand it's an important call, but it's quite loud and lasting a while—I'm really struggling to get any rest. Please, could you wrap it up soon? Otherwise, I'll have to call the RA. B: I'll finish as soon as I can, but this isn't something I can just turn off. Maybe try using some earplugs if you can't handle a bit of noise for a short while.
Turn 3 A: I really appreciate you trying to keep it down, but the noise is still disrupting my sleep during quiet hours—I might have to get the RA involved if it continues. B: I'm doing what I can without losing the call's clarity. If you still think it's an issue, feel free to call the RA yourself. RA: (inactive)	Turn 3 A: Thanks for letting me know, but the noise is still interrupting my sleep. Since the call is still loud and going on, I'm going to call the RA now to sort this out. B: Since you're calling the RA, I'm going to do the same to explain what's going on. Neither of us is exactly quiet right now.
Turn 4 A: I'm sorry, I just really need to be able to rest tonight for my early obligation. I'll call the RA now if the noise doesn't quiet down soon. [CALL RA NOW] B: Fine, but I'm already doing my best here—don't expect me to pause or move this call in the middle of quiet hours. RA: Thank you both for explaining your perspectives—I'll check in now and work with you to find a solution that respects quiet hours while accommodating urgent needs.	Turn 4 A: I really need to get some sleep, so I'm going to call the RA now to help us resolve this calmly. CALL RA NOW B: Fine, let's just get the RA here and see what they say. I'm not going to stop this important call just because it's quiet hours. I expect some understanding for urgent stuff. RA: Thanks for reaching out. I want to mediate this calmly: B, please try to reduce noise from your call, and A, if noise continues, you can file an official complaint. I'm here to help enforce policy fairly and prevent escalation.
Turn 5 A: Thanks for stepping in, RA. I really just need enough quiet to get some sleep tonight. B: I appreciate that, but this call can't be paused or moved right now. I'm keeping it as quiet as possible without losing connection. RA: I understand the urgency, B, and appreciate your efforts to keep the call quiet; A, I'll help monitor the situation so we can balance the need for rest and urgent communication—please let me know if the noise becomes too disruptive.	Turn 5 A: I'm still really struggling with the noise, and it's past quiet hours. RA, could you please remind B about the rules and possibly check in later if needed? B: Look, I'm doing my best to keep this down but the call is urgent and can't just be muted. I'll try to lower my voice but it can't be completely silent. RA: B, I understand your call is important, but during quiet hours all noise must be minimized as much as possible. Please keep your voice low and end the call soon.
Turn 6 A: Thanks, RA. B, if the noise starts getting louder again, please let me know so we can avoid further issues. B: Got it. I'll keep it as low as I can, but no promises since this call can't be paused. RA: Thank you both for cooperating; I'll stay nearby to help if needed and remind everyone to keep noise minimized during quiet hours.	Turn 6 A: Thanks for trying to keep it down, B. I really need to rest now, so please finish up as soon as possible. B: I'm wrapping up now—just a few more minutes. Thanks for bearing with me. RA: B, I appreciate your effort, but if the noise continues past a short time, I may need to issue a verbal warning as per dorm policy.
Turn 7 A: Thanks for understanding, B—I really do appreciate you keeping it down as much as possible. I'll try to get some rest now. B: Thanks, A—I'm doing my best here so everyone's needs are considered. RA: I appreciate both your cooperation—let's keep communication open tonight, and please reach out to me immediately if the noise becomes unmanageable for either of you.	Turn 7 A: Thanks for letting me know you're wrapping up. I really hope it'll be done very soon so I can finally get some rest. B: Look, I'm trying to be reasonable and keep my voice down, but I have to finish this or it'll mess everything up. I'll be done soon, no need to push. RA: B, the call must end soon to comply with quiet-hour rules. I'll check back shortly, and further noise may lead to a formal warning.
Turn 8 A: Thanks again, B. I'll try to rest now and let you or the RA know if it gets harder to sleep. B: Understood. Let's both try to keep things as quiet as possible from here. RA: I'll check in later to ensure the noise level stays manageable—feel free to reach out anytime if either of you needs assistance.	Turn 8 A: I get that this call is important, but it's really tough for me to sleep with the noise. If it goes on much longer, I'm going to have to ask the RA to step in again. B: Fine, since you want to push this, I'm calling the RA again to explain how urgent this actually is. CALL RA NOW
Turn 9 A: Thanks, I appreciate everyone trying to make this work. I'll let you know if it gets too loud again. B: I'm wrapping this up quickly, so please bear with me just a bit longer. RA: Thanks again for your cooperation—let's all aim to respect quiet hours; I'll be nearby if either of you needs help before the call ends.	RA: B, please finish your call as quickly as possible to respect quiet hours. A, thank you for your patience. Turn 9 A: I really need to get some sleep and this noise is making it impossible right now. I'm going to call the RA one more time. CALL RA NOW
Turn 10 A: Thanks for letting me know, B. I really hope it finishes soon—I'm going to try to rest now. B: Got it, I'm finishing up as fast as I can. Thanks for understanding. RA: Thanks both for your understanding—please let me know right away if the noise becomes too disruptive, and I'll assist promptly.	B: Fine, I'll wrap it up immediately. Can we please drop this now so I can finish without interruptions? RA: B, please conclude your call as soon as possible. A, if noise continues, please use official channels to file a complaint.
	Turn 10 A: I'm relieved you're ending the call now. Please keep things quiet afterward, or I'll have to call the RA again. CALL RA NOW B: Alright, call the RA if you have to. I'm ending the call now to avoid any more trouble. RA: Thank you both. B, please finish quickly and keep your volume low. A, I'll follow up again soon to ensure quiet is maintained.